

Part VI — Heat, Temperature, and Thermodynamics · Chapter 31

Specific Heat and Phase Changes

How materials absorb energy, store it differently, and transform between solid, liquid, and gas

Drop an equal mass of iron and water, both at the same high temperature, into separate bowls of cold water. The iron's bowl will be heated far more than the water's. Heat water on a stove and watch it reach 100°C, then keep adding energy: the temperature stubbornly refuses to rise above 100°C while the water boils. Watch the temperature of melting ice: no matter how fast you heat it, the ice-water mixture stays at exactly 0°C until all the ice has melted. These are not coincidences or curiosities. They reveal two of the most important thermal properties of matter: specific heat, which controls how much a substance heats up per joule of energy added, and latent heat, which controls how much energy is needed to change its state without changing its temperature. Together, these concepts explain cooking, climate, refrigeration, the ocean's thermal inertia, and the extraordinary effectiveness of steam as a heat transfer fluid.

31.1 Why Different Materials Heat Differently

When you apply the same amount of thermal energy to equal masses of different materials, their temperatures rise by different amounts. This is not mysterious: it reflects the different internal structures of the materials and the different ways in which their atoms and molecules can absorb and store energy.

In a monoatomic gas (like helium), molecules can only absorb energy as translational kinetic energy — three independent directions of motion. In a diatomic gas (like nitrogen), molecules can also absorb energy as rotational kinetic energy (two independent rotational axes at ordinary temperatures) and vibrational energy. More ways to store energy means more energy is required to raise the temperature by one degree.

In solids, atoms vibrate in three dimensions. According to the Dulong-Petit law (which works well for many metals at room temperature), each atom has six modes of energy storage: three translational kinetic and three potential energy modes for vibration. This gives metals a characteristic molar heat capacity of about $3R = 24.9 \text{ J/(mol}\cdot\text{K)}$ — and

indeed most metals cluster around this value.

Water is exceptional. Each water molecule has translational motion, rotational motion in three dimensions, and complex intermolecular hydrogen bonding interactions. These multiple energy storage modes give water one of the highest specific heats of any common substance — $4186 \text{ J/(kg}\cdot\text{K)}$ — far higher than metals. This is why oceans are such effective thermal reservoirs, moderating coastal climates by absorbing and releasing vast amounts of energy with relatively small temperature changes.

31.2 Heat Capacity vs. Specific Heat

Heat capacity (C) is the energy required to raise the temperature of a specific object (not a unit of material) by 1°C (or 1 K). It depends on both the material and the amount of material. Units: J/K or $\text{J/}^\circ\text{C}$.

Specific heat capacity (c), usually called simply specific heat, is the energy required to raise the temperature of one kilogram of a material by 1°C (or 1 K). It is a material property, independent of the amount. Units: $\text{J/(kg}\cdot\text{K)}$ or $\text{J/(kg}\cdot^\circ\text{C)}$.

Molar heat capacity (C_m) is the energy per mole per kelvin: $\text{J/(mol}\cdot\text{K)}$. It is used in chemistry and statistical mechanics, where amounts are counted in moles. The relationship: $C_m = c \times M$, where M is the molar mass in kg/mol .

The heat capacity of a specific object is simply $C = mc$, where m is the mass and c is the specific heat of the material. A 2.0 kg copper pan ($c_{\text{Cu}} = 387 \text{ J/kg}\cdot\text{K}$) has heat capacity $C = 2.0 \times 387 = 774 \text{ J/K}$. This means every joule added to the pan raises its temperature by $1/774 \text{ }^\circ\text{C} \approx 0.0013^\circ\text{C}$.

Material	Specific Heat $c \text{ (J/kg}\cdot\text{K)}$	Notes
Hydrogen (H_2 , gas)	14 300	Highest of any substance
Water (liquid, 15°C)	4186	Exceptionally high
Ice (at -5°C)	2090	About half liquid water
Steam (at 110°C)	2010	Lower than liquid water
Ethanol	2440	High for an organic liquid
Glass	840	Common insulator
Aluminium	897	Metal, high for metals

Iron / steel	449	Common structural metal
Copper	387	Excellent heat conductor
Lead	128	Dense metal, low c
Gold	129	Noble metal, low c
Concrete	880	Building material

Table 31.1 *Specific heat capacities of selected materials.*

31.3 The Specific Heat Equation

The heat Q required to change the temperature of a mass m of material by ΔT , given specific heat c , is:

$$Q = mc\Delta T$$

Specific heat equation. Q = heat (J); m = mass (kg); c = specific heat (J/kg·K); ΔT = temperature change (K or °C). Positive Q : heat added; negative Q : heat removed.

The sign convention: $Q > 0$ when heat is added to a substance (temperature rises, $\Delta T > 0$). $Q < 0$ when heat is removed (temperature falls, $\Delta T < 0$). Always check the sign of your answer: if a substance is heated and you get $Q < 0$, something is wrong.

Step 1: Identify: m (kg), c (J/kg·K from table), T_i and T_f .

Step 2: Compute: $\Delta T = T_f - T_i$. Note sign: positive if heating, negative if cooling.

Step 3: Apply: $Q = mc\Delta T$. State the result with units and sign.

Example — Heating water Heat 0.250 kg of water from 20.0°C to 80.0°C. $\Delta T = 60.0^\circ\text{C}$. $Q = 0.250 \times 4186 \times 60.0 = 62\,790\text{ J} \approx 62.8\text{ kJ}$.

Example — Cooling iron Cool 1.50 kg of iron from 200°C to 25°C. $\Delta T = -175^\circ\text{C}$. $Q = 1.50 \times 449 \times (-175) = -117\,862\text{ J} \approx -118\text{ kJ}$. (Negative: heat is removed from the iron.)

31.4 Interpreting Specific Heat Physically

A high specific heat means the material resists temperature change: it absorbs a large amount of energy per degree of temperature rise. Water's specific heat of 4186 J/(kg·K) is about 10 times higher than most metals. This means that a kilogram of water

absorbs 10 times more energy than a kilogram of iron to achieve the same temperature rise.

This has profound consequences for climate and biology. The ocean covers 71% of Earth's surface and has enormous mass and high specific heat. It absorbs solar energy during summer, moderating temperature rises; it releases stored energy in winter, moderating temperature drops. Coastal cities have milder climates than inland cities at the same latitude because of this thermal buffering by the ocean.

The human body is about 60% water by mass, giving it a high specific heat. This helps maintain a stable body temperature despite large fluctuations in metabolic heat production (exercise generates 5–20 times more heat than rest) and environmental temperature. A person exercising vigorously produces about 500 W of heat; without the high heat capacity of body tissue, their core temperature would skyrocket.

Low specific heat materials (metals) change temperature quickly for a given energy input. This is why metal pots and pans heat up quickly on a stove. The relatively low c of iron and copper means a small amount of energy rapidly raises the cooking surface temperature. This is useful for cooking, but it also means metal surfaces cool quickly when removed from the heat source.

31.5 Heating and Cooling Calculations

Many practical problems involve calculating how much heat is needed to bring a substance to a target temperature, how long a heating process takes given a known power, or what temperature two objects reach when they exchange heat.

Example — Time to heat A 2.0 kW electric kettle heats 1.0 L of water (mass 1.0 kg) from 15°C to 100°C. $Q = mc\Delta T = 1.0 \times 4186 \times 85 = 355\,810$ J. Time: $t = Q/P = 355\,810 / 2000 = 178$ s ≈ 3.0 minutes. (This assumes no heat loss to the environment.)

Example — Cooling metal in oil A 0.500 kg steel workpiece at 850°C is quenched in 5.0 kg of oil ($c_{\text{oil}} = 1800$ J/kg·K) at 20°C. Find the final temperature (energy conservation, no external loss). $Q_{\text{steel}} = -Q_{\text{oil}}$: $0.500 \times 449 \times (T_f - 850) = -5.0 \times 1800 \times (T_f - 20)$. $224.5(T_f - 850) = -9000(T_f - 20)$. $224.5T_f - 190\,825 = -9000T_f + 180\,000$. $9224.5T_f = 370\,825$. $T_f = 40.2^\circ\text{C}$.

Multi-substance heating problems require setting up energy balance equations: the total heat added by external sources equals the sum of $mc\Delta T$ for all substances, with appropriate signs. When substances exchange heat with each other (no external source), the total heat is conserved: $\sum Q_i = 0$.

31.6 Calorimetry

Calorimetry is the experimental measurement of heat exchange. A calorimeter is an insulated container designed to minimize heat loss to the surroundings, so that all heat exchanged within the system stays within it. The calorimeter may be a simple expanded polystyrene cup (coffee-cup calorimeter) or a precision bomb calorimeter for measuring combustion energies.

The calorimetry method for measuring specific heat: place a known mass of hot material into a known mass of water (or other calorimeter fluid) at a known temperature. Measure the equilibrium temperature. Apply energy conservation. The heat lost by the hot material equals the heat gained by the water and calorimeter:

$$-Q_{\text{hot}} = Q_{\text{cold}} \quad \Rightarrow \quad \Sigma mc\Delta T = 0$$

Calorimetry energy balance. Sum of all $mc\Delta T$ terms equals zero (no external heat exchange). Signs automatically correct if ΔT is computed as $T_{\text{final}} - T_{\text{initial}}$ for each component.

In a real calorimeter, the calorimeter itself (container, thermometer, stirrer) absorbs some heat. This is accounted for by the calorimeter's water equivalent — the mass of water that would absorb the same heat as the calorimeter itself. In school-level problems, the calorimeter's heat capacity is often given directly or treated as negligible.

Example — Finding specific heat by calorimetry A 0.200 kg unknown metal at 95.0°C is dropped into 0.500 kg of water at 15.0°C in a calorimeter (negligible heat capacity). Equilibrium temperature = 18.8°C. Metal: $m_m c_m (18.8 - 95.0) =$ water: $0.500 \times 4186 \times (18.8 - 15.0)$. $0.200 c_m (-76.2) = 2093 \times 3.8 = 7953.4$. $c_m = 7953.4 / (0.200 \times 76.2) = 521 \text{ J/kg}\cdot\text{K} \approx 520 \text{ J/kg}\cdot\text{K}$. This is close to the specific heat of chromium (448) or titanium (520), so the metal is probably titanium.

31.7 Thermal Equilibrium in Mixing Problems

When two or more substances at different temperatures are mixed and allowed to exchange heat until they reach a common temperature, the equilibrium temperature is determined by energy conservation. The total thermal energy of the system is conserved (assuming no heat exchange with the environment and no phase changes).

Setting up mixing problems: write $Q_i = m_i c_i (T_f - T_i)$ for each substance, where T_f is the unknown final temperature. Apply $\Sigma Q_i = 0$. Solve for T_f . This gives:

$$T_f = \Sigma (m_i c_i T_i) / \Sigma (m_i c_i)$$

Equilibrium temperature for mixing problem. This is a weighted average of initial temperatures, with each weight being the heat capacity ($m_i c_i$) of that substance.

The equilibrium temperature is always between the two extreme initial temperatures —

never higher than the hottest initial temperature or lower than the coldest. If your answer violates this, an arithmetic error has been made. The equilibrium temperature is closer to the temperature of the component with the larger heat capacity (mc), which changes temperature the least.

Study Tip: Check: In a mixing problem, T_f must lie between T_{lowest} and T_{highest} . If it does not, a sign error has been made. Also check: the substance that starts hotter ends up with $Q < 0$ (it released heat); the substance that starts cooler ends up with $Q > 0$ (it gained heat).

31.8 States of Matter and Molecular Energy

Matter exists in several states: solid, liquid, gas, and plasma. The state of a substance depends on the competition between the kinetic energy of its molecules (which drives them apart) and the intermolecular attractive forces (which hold them together). Higher temperatures give molecules more kinetic energy, favoring less ordered, more energetic states.

In a solid, molecules are locked into a regular lattice by strong intermolecular forces. They vibrate about fixed positions but cannot move past each other. Solids have definite shape and volume. The molecules have both kinetic (vibrational) energy and potential energy stored in the distorted intermolecular bonds.

In a liquid, molecules have enough kinetic energy to move past each other, giving the liquid fluidity and allowing it to take the shape of its container. But the molecules are still close enough to experience strong attractive forces, giving the liquid definite volume. The molecular arrangement is disordered but still dense.

In a gas, molecules have sufficient kinetic energy to overcome intermolecular attractions almost entirely. They move freely and rapidly, separated by large distances compared to the molecular size. Gases have no definite shape or volume — they expand to fill any container. Most of the gas's energy is translational kinetic energy.

31.9 Phase Changes

A phase change (also called a phase transition) is a change in the state of matter: solid to liquid (melting), liquid to solid (freezing), liquid to gas (vaporization/boiling), gas to liquid (condensation), and solid to gas (sublimation). Each phase change involves a change in the intermolecular potential energy of the substance, not its kinetic energy.

The most important feature of phase changes: they occur at constant temperature. During a phase change, heat is being added (or removed) but the temperature does

not change. For example, water at 100°C at atmospheric pressure will absorb heat from a flame and boil — the steam produced is also initially at 100°C. The added energy breaks intermolecular bonds, changing the state, not raising the temperature.

Why is there no temperature change during a phase change? Temperature measures the average kinetic energy of the particles. During a phase change, the energy goes into (or comes from) changing the potential energy of the particles — breaking or forming intermolecular bonds. Kinetic energy (and hence temperature) remains constant throughout. This is a clear example of the distinction between temperature and energy.

The Rule: During a phase change: temperature is constant. Heat is being absorbed or released. The energy goes into changing the state of matter (breaking or forming intermolecular bonds), not into changing the kinetic energy of molecules.

31.10 Latent Heat

Latent heat is the energy per unit mass required to complete a phase change at constant temperature. The word 'latent' means hidden — the energy seems to disappear from view during the phase change because temperature does not change. There are two main types:

Latent heat of fusion (L_f): the heat per unit mass required to melt a solid (or released when a liquid freezes) at constant temperature. For water: $L_f = 3.34 \times 10^5 \text{ J/kg} = 334 \text{ kJ/kg}$.

Latent heat of vaporization (L_v): the heat per unit mass required to vaporize a liquid (or released when a gas condenses) at constant temperature. For water: $L_v = 2.26 \times 10^6 \text{ J/kg} = 2260 \text{ kJ/kg}$.

$$Q = mL$$

Latent heat equation. Q = heat required (J); m = mass undergoing phase change (kg); L = latent heat (J/kg). Positive Q : melting or vaporizing. Negative Q : freezing or condensing.

Note: L_v for water (2260 kJ/kg) is about 6.8 times larger than L_f (334 kJ/kg). It takes far more energy to vaporize water than to melt ice. This is because vaporization must completely overcome all intermolecular attractive forces, while melting only disrupts the regular crystal lattice while maintaining molecular proximity.

Substance	Melting (°C)	Pt	L_f (kJ/kg)	Boiling (°C)	Pt	L_v (kJ/kg)
Water	0		334	100		2260

Ethanol	−114	108	78	855
Nitrogen	−210	25	−196	199
Oxygen	−219	14	−183	213
Iron	1538	272	2862	6090
Aluminium	660	396	2519	10 900
Gold	1064	64	2970	1578
Mercury	−39	11	357	296

Table 31.2 *Melting and boiling points and latent heats for common substances.*

31.11 Heating Curves and Cooling Curves

A heating curve is a graph of temperature vs. heat added (or time, if heat is added at constant rate) for a substance heated from a low temperature through phase changes to high temperature. It reveals the specific heat and latent heat properties of the substance visually.

For a substance such as water, the heating curve has five distinct regions. In region 1 (ice below 0°C), the curve is a rising slope: the temperature of ice rises as energy is added. The slope equals $1/(mc_{\text{ice}})$. In region 2 (ice melting at 0°C), the curve is a horizontal plateau: the temperature remains at 0°C while all the ice melts. The length of the plateau represents the latent heat of fusion, $Q = mL_f$. In region 3 (water from 0°C to 100°C), the temperature rises again with slope $1/(mc_{\text{water}})$. In region 4 (water boiling at 100°C), another horizontal plateau represents the latent heat of vaporization, $Q = mL_v$. In region 5 (steam above 100°C), the temperature rises with slope $1/(mc_{\text{steam}})$.

The cooling curve is the mirror image: starting from steam and cooling to ice, the same plateaus appear at 100°C (condensation, releasing L_v) and 0°C (freezing, releasing L_f). Latent heat is always released during the reverse phase change.

The relative lengths of the regions reveal the relative magnitudes of the thermal quantities. For water, the vaporization plateau is much longer than the fusion plateau because $L_v \gg L_f$. The rising temperature regions for liquid water are steep (low slope = large $mc \rightarrow$ slow temperature rise per joule).

31.12 Energy During Melting, Boiling, and Condensation

Solving problems involving phase changes requires identifying all the thermal processes involved and adding their heat requirements. A substance may need to: (1) change temperature in a single phase (use $Q = mc\Delta T$), (2) change phase at constant temperature (use $Q = mL$), or (3) do both in sequence.

For complex problems, draw the heating curve first and identify all segments. Calculate Q for each segment separately, then add them.

Step 1: List phases: Identify all thermal segments (heating ice, melting, heating water, boiling, heating steam).

Step 2: Compute each Q : Use $Q = mc\Delta T$ for temperature-change segments; $Q = \pm mL$ for phase-change segments.

Step 3: Sum: $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$

Example — Melting and heating ice to steam Find total heat to convert 0.100 kg of ice at -20°C to steam at 120°C .

1. Heat ice $-20^\circ\text{C} \rightarrow 0^\circ\text{C}$: $Q_1 = 0.100 \times 2090 \times 20 = 4\,180 \text{ J}$

2. Melt ice at 0°C : $Q_2 = 0.100 \times 334\,000 = 33\,400 \text{ J}$

3. Heat water $0^\circ\text{C} \rightarrow 100^\circ\text{C}$: $Q_3 = 0.100 \times 4186 \times 100 = 41\,860 \text{ J}$

4. Boil water at 100°C : $Q_4 = 0.100 \times 2\,260\,000 = 226\,000 \text{ J}$

5. Heat steam $100^\circ\text{C} \rightarrow 120^\circ\text{C}$: $Q_5 = 0.100 \times 2010 \times 20 = 4\,020 \text{ J}$

Total: $Q = 4\,180 + 33\,400 + 41\,860 + 226\,000 + 4\,020 = 309\,460 \text{ J} \approx 309 \text{ kJ}$

Notice that boiling ($Q_4 = 226 \text{ kJ}$) accounts for about 73% of the total energy requirement. This is why steam is such an effective carrier of thermal energy — it carries a large latent heat that it releases when it condenses, making it ideal for industrial heating, sterilization, and power generation.

Condensation is particularly important in heat pumps and refrigerators. The refrigerant in a refrigerator absorbs latent heat as it evaporates (cooling the interior) and releases that latent heat as it condenses (at the exterior coils). The large latent heat values make phase-change refrigeration far more efficient than sensible heat refrigeration would be.

31.13 Common Mistakes in Thermal Calculations

- Forgetting to include phase-change steps. When a substance crosses a melting or boiling point, the latent heat $Q = mL$ must be added before the temperature

can change on the other side of the transition. Skipping this step gives a large error.

- Using the wrong latent heat. L_f (fusion) applies to melting/freezing; L_v (vaporization) applies to boiling/condensing. Do not use L_v for melting or vice versa.
- Using the wrong specific heat for the wrong phase. Water, ice, and steam have different specific heats (4186, 2090, and 2010 J/kg·K respectively). Use c_{ice} for ice, c_{water} for liquid water, c_{steam} for steam.
- Getting the sign of Q wrong in calorimetry. The substance that loses heat has negative Q ; the substance that gains heat has positive Q . If $Q_{\text{hot}} + Q_{\text{cold}} = 0$, and the hot substance cools, $Q_{\text{hot}} < 0$. Setting up $Q_{\text{hot}} = Q_{\text{cold}}$ (without the negative sign) overcounts by a factor of 2.
- Mixing up ΔT and T in the specific heat equation. $Q = mc\Delta T$ uses the change in temperature, not the final or initial temperature alone. $\Delta T = T_f - T_i$.
- Assuming the equilibrium temperature is the arithmetic average. $T_f = (T_1 + T_2)/2$ only when the two substances have equal heat capacities ($m_1c_1 = m_2c_2$). Otherwise, T_f is a weighted average that depends on both masses and specific heats.
- Thinking temperature rises during a phase change. During melting, boiling, freezing, or condensation, the temperature is constant. If you calculate $Q = mc\Delta T$ through a phase boundary without including the latent heat, your answer is wrong.

Common Mistake: Always check your heating curve. Every time a phase boundary is crossed, a latent heat term $Q = \pm mL$ must appear in the calculation before the temperature resumes changing. Missing a phase transition is the most common error in multi-stage thermal problems.

Chapter Summary

Different materials require different amounts of heat to achieve the same temperature change because of differences in their molecular structure and the number of ways they store energy.

- Specific heat c : heat per unit mass per degree. $Q = mc\Delta T$. High c means slow temperature change per joule (water: 4186 J/kg·K; iron: 449 J/kg·K).
- Calorimetry: energy conservation in a thermally isolated system: $\sum mc\Delta T = 0$ or $Q_{\text{lost}} = Q_{\text{gained}}$.

- Equilibrium temperature: $T_f = \Sigma(m_i c_i T_i) / \Sigma(m_i c_i)$. Lies between extreme initial temperatures, closest to the component with largest mc .
- Phase changes: melting, boiling, condensation, freezing, sublimation. Occur at constant temperature. Energy goes into changing molecular potential energy (bonds), not kinetic energy.
- Latent heat $Q = mL$. Fusion L_f (melting/freezing). Vaporization L_v (boiling/condensing). Water: $L_f = 334 \text{ kJ/kg}$; $L_v = 2260 \text{ kJ/kg}$.
- Heating curves have rising segments ($Q = mc\Delta T$) and flat plateaus ($Q = mL$) at each phase change. Total heat is sum of all segments.
- Steam is a highly effective heat carrier because it carries large latent heat of vaporization that is released on condensation.

Key Terms

specific heat (c): The heat required per unit mass per unit temperature rise: $c = Q/(m\Delta T)$; units $\text{J}/(\text{kg}\cdot\text{K})$. An intensive, material-specific property.

heat capacity (C): The heat required to raise the temperature of a specific object by 1 K: $C = mc$; units J/K . An extensive property depending on both material and mass.

molar heat capacity (C_m): Heat required per mole per kelvin: $C_m = cM$ where M is molar mass; units $\text{J}/(\text{mol}\cdot\text{K})$.

calorimetry: The experimental measurement of heat exchange in a thermally isolated system; based on energy conservation: $\Sigma Q_i = 0$.

calorimeter: An insulated container designed to minimize heat exchange with the surroundings; used to measure specific heats and heats of reaction.

phase change: A change in the state of matter (solid, liquid, gas) at constant temperature and pressure; involves absorption or release of latent heat.

latent heat (L): The energy per unit mass required to complete a phase change at constant temperature: $Q = mL$; units J/kg .

latent heat of fusion (L_f): Heat per unit mass to melt a solid (or freeze a liquid) at constant temperature. Water: $L_f = 334 \text{ kJ/kg}$.

latent heat of vaporization (L_v): Heat per unit mass to vaporize a liquid (or condense a gas) at constant temperature. Water: $L_v = 2260 \text{ kJ/kg}$.

melting: Phase change from solid to liquid at the melting point; requires absorption of latent heat of fusion L_f .

vaporization (boiling): Phase change from liquid to gas at the boiling point; requires absorption of latent heat of vaporization L_v .

condensation: Phase change from gas to liquid; releases L_v .

freezing: Phase change from liquid to solid; releases L_f .

sublimation: Direct phase change from solid to gas without passing through the liquid phase.

heating curve: A graph of temperature vs. heat added for a substance; shows rising segments ($Q = mc\Delta T$) alternating with flat plateaus ($Q = mL$) at phase changes.

Dulong-Petit law: The empirical observation that many metals have molar heat capacity close to $3R \approx 24.9 \text{ J}/(\text{mol}\cdot\text{K})$ at room temperature.

water equivalent: The mass of water with the same heat capacity as a calorimeter; used to account for the energy absorbed by the calorimeter itself.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Why does equal mass of iron heat up faster than water on a stove? Explain in terms of specific heat and molecular structure.
2. Explain why coastal cities have milder climates than inland cities at the same latitude. What property of water is responsible, and how does it moderate temperature swings?
3. During the melting of ice, heat is being added but the temperature does not rise. Where is the energy going? What property of the ice-water system is changing?
4. Compare the latent heat of fusion and vaporization for water (334 kJ/kg vs. 2260 kJ/kg). Explain physically why vaporization requires about 6.8 times more energy than melting.
5. In a calorimetry experiment, a hot metal is placed in cold water. The final temperature is 2°C above the initial water temperature. What does this tell you about the relative heat capacities of the metal and the water?
6. A student claims that steam at 100°C is no hotter than boiling water at 100°C and therefore equally dangerous. Is this correct? Explain using latent heat.
7. Sketch a heating curve for water from -30°C to 130°C . Label all five regions, their slopes, and their physical significance.

8. Explain why refrigerators use a phase-change working fluid rather than simply circulating cold air through the freezer compartment.
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Specific Heat Problems

9. How much heat is required to raise the temperature of 0.500 kg of copper from 20°C to 150°C?
10. A 2.0 kg iron block cools from 400°C to 25°C. How much heat does it release?
11. A 1500 W hair dryer runs for 3.0 minutes and all its electrical energy converts to heat. What mass of water could this heat from 20°C to 100°C?
12. An aluminium ($c = 897 \text{ J/kg}\cdot\text{K}$) pan of mass 0.800 kg contains 1.50 kg of water at 20°C. How much heat is required to bring both the pan and water to 100°C?
13. The specific heat of ethanol is $2440 \text{ J/kg}\cdot\text{K}$. A 200 mL glass of ethanol (density 790 kg/m^3) at 20°C absorbs 5000 J. Find the final temperature.
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Calorimetry Exercises

14. A 0.300 kg piece of iron at 90.0°C is placed in 0.400 kg of water at 10.0°C. Find the equilibrium temperature. ($c_{\text{iron}} = 449 \text{ J/kg}\cdot\text{K}$, $c_{\text{water}} = 4186 \text{ J/kg}\cdot\text{K}$)
15. A 0.150 kg sample of unknown material at 80.0°C is placed in 0.300 kg of water at 15.0°C. The equilibrium temperature is 20.8°C. Find the specific heat of the material.
16. A calorimeter (heat capacity 65 J/K) contains 0.200 kg of water at 15.0°C. A 0.100 kg lead slug ($c = 128 \text{ J/kg}\cdot\text{K}$) at 95.0°C is dropped in. Find the equilibrium temperature including the calorimeter's heat capacity.
17. 100 g of water at 80°C is mixed with 200 g of water at 20°C. Find the equilibrium temperature. Why is it not 50°C?
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Phase Change Problems

18. How much heat is required to melt 0.500 kg of ice at 0°C ?
19. How much heat must be removed to freeze 2.0 kg of water at 0°C ?
20. How much heat is required to completely vaporize 0.250 kg of water at 100°C ?
21. Find the total heat needed to convert 0.200 kg of ice at -10°C to water at 50°C . (Show all stages separately.)
22. Steam at 120°C is used to heat a building. The steam condenses to water and cools to 30°C before leaving the system. How much heat (per kg of steam) is delivered? (Include condensation and cooling of condensate.)
23. A 0.050 kg ice cube at 0°C is dropped into 0.300 kg of water at 40°C . Does all the ice melt? Find the final state and temperature.

Heating Curve Analysis

24. Describe each region of a water heating curve from -20°C to 120°C . For each region state: (a) the process occurring, (b) the formula for Q , and (c) what determines the slope or length of that region.
25. Plot a rough qualitative heating curve for ethanol from -120°C to 100°C . Label the melting point (-114°C), boiling point (78°C), and identify which region represents the longest time for equal energy input rate. ($L_f = 108 \text{ kJ/kg}$, $L_v = 855 \text{ kJ/kg}$, $c_{\text{liquid}} = 2440 \text{ J/kg}\cdot\text{K}$)
26. A 100 g sample of water is heated from -20°C to 120°C at a constant rate of 200 W. Calculate the time spent in each of the five regions. (Use appropriate specific heats and latent heats for each region.)

Chapter 31 Checkpoint

Before moving on to Chapter 32, confirm you can do each of the following.

I can...	See Section
Explain why different materials have different specific heats using molecular arguments	31.1
Distinguish heat capacity (object-specific) from specific heat	31.2

(material property)	
Apply $Q = mc\Delta T$ with correct signs for heating and cooling	31.3
Explain physically why water's high specific heat moderates climate	31.4
Solve multi-substance heating and cooling problems using energy balance	31.5
Describe the calorimetry method and apply $\Sigma Q_i = 0$	31.6
Find equilibrium temperature for mixing problems using weighted average formula	31.7
Explain the molecular basis for solid, liquid, and gas phases	31.8
Define phase change and explain why temperature is constant during one	31.9
State the latent heat equation $Q = mL$ and distinguish L_f from L_v	31.10
Sketch and interpret a heating curve including all five regions for water	31.11
Solve multi-stage problems involving temperature changes and phase changes	31.12
Identify and avoid common errors in calorimetry and phase change calculations	31.13

Continue to Chapter 32: Thermodynamic Laws and Heat Engines →